

Mathematical Foundations of Political Methodology

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- Determinant
- Eigenvalues and Eigenvectors
- Multivariate Function
- Multivariate Derivative

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Multivariable Calculus

Functions of many variables:

- 1) Policies may be multidimensional (policy provision and pork buy off)
- 2) Countries may invest in offensive and defensive resources for fighting wars
- 3) Ethnicity and resources could affect investment

Today:

- 0) Determinant
- 0) Eigenvector/Diagonalization
- 1) Multivariate functions
- 2) Partial Derivatives, Gradients, Jacobians, and Hessians
- 3) Total Derivative, Implicit Differentiation, Implicit Function Theorem
- 4) Multivariate Integration

Determinant

Suppose we have a **square** ($n \times n$) matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

A determinant is a function that assigns a number to square matrices

Determinant

Facts needed to define determinant :

Definition

A *permutation* of the set of integers $\{1, 2, \dots, J\}$ is an arrangement of these integers in some order without omissions or repetition.

For example, consider $\{1, 2, 3, 4\}$

$\{3, 2, 1, 4\}$

$\{4, 3, 2, 1\}$

If we have J integers then there are $J!$ permutations

Determinant

Definition

An *inversion* occurs when a larger integer occurs before a smaller integer in a permutation

Even permutation: total inversions are even

Odd permutation: total inversions are odd

Count the inversions

$\{3, 2, 1\}$

$\{1, 2, 3\}$

$\{3, 1, 2\}$

$\{2, 1, 3\}$

$\{1, 3, 2\}$

$\{2, 3, 1\}$

Determinant

Definition

*For a square $n \times n$ matrix A , we will call an **elementary product** an n element long product, with no two components coming from the same row or column. We will call a **signed** elementary product one that multiplies odd permutations of the column numbers by -1 .*

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$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

There are $n!$ elementary products

Determinant

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*Suppose A is an $n \times n$ matrix. Define the determinant function $\det(A)$ to be the sum of signed elementary products from A . Call $\det(A)$ the **determinant** of A*

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$$\begin{aligned}\det(A) &= \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \\ &= a_{11}a_{22} - a_{12}a_{21}\end{aligned}$$

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R Code!

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- Multivariate Function
- Multivariate Derivative

An Introduction to Eigenvectors, Values, and Diagonalization

Definition

Suppose $\mathbf{A}$ is an $N \times N$ matrix and λ is a scalar.
If

$$\mathbf{Ax} = \lambda \mathbf{x}$$

Then $\mathbf{x}$ is an *eigenvector* and λ is the associated *eigenvalue*

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 - Find vectors in *null space* of:

$$(\mathbf{A} - \lambda \mathbf{I}) = 0$$

An Introduction to Eigenvectors, Values, and Diagonalization

Theorem

Suppose $\mathbf{A}$ is an *invertible* $N \times N$ matrix and further suppose that $\mathbf{A}$ has N distinct eigenvalues and N linearly independent eigenvectors. Then we can write $\mathbf{A}$ as,

$$\mathbf{A} = \mathbf{W} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{pmatrix} \mathbf{W}^{-1}$$

where $\mathbf{W} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_N)$ is an $N \times N$ matrix with the N eigenvectors as column vectors.

Proof:
Note

$$\begin{aligned}\mathbf{A}\mathbf{W} &= (\lambda_1 \mathbf{w}_1 \quad \lambda_2 \mathbf{w}_2 \quad \dots \quad \lambda_N \mathbf{w}_N) \\ &= \mathbf{W}\mathbf{\Lambda} \\ \mathbf{A} &= \mathbf{W}\mathbf{\Lambda}\mathbf{W}^{-1}\end{aligned}$$

Examples of Diagonalization

Suppose $\mathbf{A}$ is an $N \times N$ invertible matrix with eigenvalues $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_N)$ and eigenvectors $\mathbf{W}$. Calculate $\mathbf{A}\mathbf{A} = \mathbf{A}^2$

$$\begin{aligned}\mathbf{A}\mathbf{A} &= \mathbf{W}\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{W}\mathbf{\Lambda}\mathbf{W}^{-1} \\ &= \mathbf{W} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{pmatrix} \mathbf{W}^{-1} \\ &= \mathbf{W} \begin{pmatrix} \lambda_1^2 & 0 & \dots & 0 \\ 0 & \lambda_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N^2 \end{pmatrix} \mathbf{W}^{-1}\end{aligned}$$

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Multivariate Functions

$$f(x_1, x_2) = x_1 + x_2$$

Multivariate Functions

$$f(x_1, x_2) = x_1^2 + x_2^2$$

Multivariate Functions

$$f(x_1, x_2) = \sin(x_1) \cos(x_2)$$

Multivariate Functions

$$f(x_1, x_2) = -(x - 5)^2 - (y - 2)^2$$

Multivariate Functions

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Multivariate Functions

$$\begin{aligned} f(\mathbf{x}) &= f(x_1, x_2, \dots, x_N) \\ &= x_1 + x_2 + \dots + x_N \\ &= \sum_{i=1}^N x_i \end{aligned}$$

Multivariate Functions

Definition

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^1$. We will call f a **multivariate** function. We will commonly write,

$$f(\mathbf{x}) = f(x_1, x_2, x_3, \dots, x_n)$$

- $\mathbb{R}^n = \mathbb{R} \underbrace{\times}_{\text{cartesian}} \mathbb{R} \times \mathbb{R} \times \dots \mathbb{R}$
- The function we consider will take n inputs and output a single number (that lives in $\mathbb{R}^1$, or the real line)

Example 1

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Evaluate at $\mathbf{x} = (x_1, x_2, x_3) = (2, 3, 2)$

$$\begin{aligned} f(2, 3, 2) &= 2 + 3 + 2 \\ &= 7 \end{aligned}$$

Example 1

$$f(x_1, x_2) = x_1 + x_2 + x_1 x_2$$

Evaluate at $\mathbf{w} = (w_1, w_2) = (1, 2)$

$$\begin{aligned} f(w_1, w_2) &= w_1 + w_2 + w_1 w_2 \\ &= 1 + 2 + 1 \times 2 \\ &= 5 \end{aligned}$$

Preferences for Multidimensional Policy

Recall that in the **spatial** model, we suppose policy and political actors are located in a space.

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$$U(\mathbf{m}) = U(1, 1, \dots, 1)$$

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Recall that in the **spatial** model, we suppose policy and political actors are located in a space.

Suppose that policy is N dimensional—or $\mathbf{x} \in \Re^N$.

Suppose that legislator i 's utility is a $U : \Re^N \rightarrow \Re^1$ and is given by,

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$$\begin{aligned}U(\mathbf{m}) &= U(1, 1, \dots, 1) \\&= -(1 - 0)^2 - (1 - 0)^2 - \dots - (1 - 0)^2 \\&= -\sum_{j=1}^N 1 = -N\end{aligned}$$

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Multivariate Derivative

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, where $X \subset \mathbb{R}^n$. $f(\mathbf{x}) = f(x_1, x_2, \dots, x_N)$. If the limit,

$$\begin{aligned}\frac{\partial}{\partial x_i} f(\mathbf{x}_0) &= \frac{\partial}{\partial x_i} f(x_{01}, x_{02}, \dots, x_{0i}, x_{0i+1}, \dots, x_{0N}) \\ &= \lim_{h \rightarrow 0} \frac{f(x_{01}, x_{02}, \dots, x_{0i} + h, \dots, x_{0N}) - f(x_{01}, x_{02}, \dots, x_{0i}, \dots, x_{0N})}{h}\end{aligned}$$

exists then we call this the partial derivative of f with respect to x_i at the value $\mathbf{x}_0 = (x_{01}, x_{02}, \dots, x_{0N})$.

Rules for Taking Partial Derivatives

Partial Derivative: $\frac{\partial f(\mathbf{x})}{\partial x_i}$

- Treat each instance of x_i as a **variable** that we would differentiate before
- Treat each instance of $\mathbf{x}_{-i} = (x_1, x_2, x_3, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ as a **constant**

Example Partial Derivatives

$$\begin{aligned} f(\mathbf{x}) &= f(x_1, x_2) \\ &= x_1 + x_2 \end{aligned}$$

Partial derivative, with respect to x_1 at (x_{01}, x_{02})

$$\begin{aligned} \frac{\partial f(x_1, x_2)}{\partial x_1} \Big|_{(x_{01}, x_{02})} &= 1 + 0 \Big|_{x_{01}, x_{02}} \\ &= 1 \end{aligned}$$

Example Partial Derivatives

$$\begin{aligned} f(\mathbf{x}) &= f(x_1, x_2, x_3) \\ &= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2 \end{aligned}$$

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What is the partial derivative with respect to x_1 ?
 $\mathbf{x}_0 = (x_{01}, x_{02}, x_{03})$.

Evaluated at

$$\begin{aligned}\frac{\partial f(\mathbf{x})}{\partial x_1} \Big|_{\mathbf{x}_0} &= 2x_1 \log(x_1) + x_1^2 \frac{1}{x_1} + x_2 x_3 \Big|_{\mathbf{x}_0} \\&= 2x_{01} \log(x_{01}) + x_{01} + x_{02} x_{03}\end{aligned}$$

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$$\begin{aligned}f(\mathbf{x}) &= f(x_1, x_2, x_3) \\&= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2\end{aligned}$$

What is the partial derivative with respect to x_1 ? x_2 ? Evaluated at $\mathbf{x}_0 = (x_{01}, x_{02}, x_{03})$.

$$\begin{aligned}\frac{\partial f(\mathbf{x})}{\partial x_2} \Big|_{\mathbf{x}_0} &= x_1 x_3 \Big|_{\mathbf{x}_0} \\&= x_{01} x_{03}\end{aligned}$$

Example Partial Derivatives

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What is the partial derivative with respect to x_1 ? x_2 ? x_3 ? Evaluated at $\mathbf{x}_0 = (x_{01}, x_{02}, x_{03})$.

$$\begin{aligned}\frac{\partial f(\mathbf{x})}{\partial x_3} \Big|_{\mathbf{x}_0} &= x_1 x_2 + 2x_3 \Big|_{\mathbf{x}_0} \\&= x_{01} x_{02} + 2x_{03}\end{aligned}$$

Rate of Change, Linear Model

Suppose we regress **Approval**_{*i*} rate for Obama in month *i* on Employ_{*i*} and Gas_{*i*}. We obtain the following model:

$$\text{Approval}_i = 0.8 - 0.5\text{Employ}_i - 0.25\text{Gas}_i$$

We are modeling $\text{Approval}_i = f(\text{Employ}_i, \text{Gas}_i)$. What is partial derivative with respect to employment?

$$\frac{\partial f(\text{Employ}_i, \text{Gas}_i)}{\partial \text{Employ}_i} = -0.5$$

Gradient

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$ with $X \subset \mathbb{R}^n$ is a differentiable function. Define the gradient vector of f at $\mathbf{x}_0$, $\nabla f(\mathbf{x}_0)$ as,

$$\nabla f(\mathbf{x}_0) = \left(\frac{\partial f(\mathbf{x}_0)}{\partial x_1}, \frac{\partial f(\mathbf{x}_0)}{\partial x_2}, \frac{\partial f(\mathbf{x}_0)}{\partial x_3}, \dots, \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \right)$$

- The gradient points in the direction that the function is **increasing** in the fastest direction
- We'll use this to do optimization (both analytic and computational)

Example Gradient Calculation

Suppose

$$\begin{aligned}f(\mathbf{x}) &= f(x_1, x_2, \dots, x_n) \\&= x_1^2 + x_2^2 + \dots + x_n^2 \\&= \sum_{i=1}^n x_i^2\end{aligned}$$

Then $\nabla f(\mathbf{x}^*)$ is

$$\nabla f(\mathbf{x}^*) = (2x_1^*, 2x_2^*, \dots, 2x_n^*)$$

So if $\mathbf{x}^* = (3, 3, \dots, 3)$ then

$$\begin{aligned}\nabla f(\mathbf{x}^*) &= (2 * 3, 2 * 3, \dots, 2 * 3) \\&= (6, 6, \dots, 6)\end{aligned}$$

Second Partial Derivative

Definition

Suppose $f : X \rightarrow \mathbb{R}$ where $X \subset \mathbb{R}^n$ and suppose that $\frac{\partial f(x_1, x_2, \dots, x_n)}{\partial x_i}$ exists. Then we define,

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} \equiv \frac{\partial}{\partial x_j} \left(\frac{\partial f(\mathbf{x})}{\partial x_i} \right)$$

- Second derivative could be with respect to x_i or with some other variable x_j
- Nagging question: does order matter?

Second Partial Derivative: Order Doesn't Matter

Theorem

Young's Theorem Let $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^n$ be a twice differentiable function on all of X . Then for any i, j , at all $\mathbf{x}^* \in X$,

$$\frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x}^*) = \frac{\partial^2}{\partial x_j \partial x_i} f(\mathbf{x}^*)$$

Second Order Partial Derivates

$$f(\mathbf{x}) = x_1^2 x_2^2$$

Then,

$$\frac{\partial^2}{\partial x_1 \partial x_1} f(\mathbf{x}) = 2x_2^2$$

$$\frac{\partial^2}{\partial x_1 \partial x_2} f(\mathbf{x}) = 4x_1 x_2$$

$$\frac{\partial^2}{\partial x_2 \partial x_2} f(\mathbf{x}) = 2x_1^2$$

Hessians

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$, with f a twice differentiable function. We will define the **Hessian** matrix as the matrix of second derivatives at $\mathbf{x}^* \in X$,

$$\mathbf{H}(f)(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

- Hessians are **symmetric**
- They describe **curvature** of a function (think, how bended)
- Will be the basis for second derivative test for multivariate optimization

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$$\mathbf{H}(f)(\mathbf{x}) = \begin{pmatrix} 2x_2^2 x_3^2 & 4x_1 x_2 x_3^2 & 4x_1 x_2^2 x_3 \\ 4x_1 x_2 x_3^2 & 2x_1^2 x_3^2 & 4x_1^2 x_2 x_3 \\ 4x_1 x_2^2 x_3 & 4x_1^2 x_2 x_3 & 2x_1^2 x_2^2 \end{pmatrix}$$

Functions with Multidimensional Codomains

Definition

Suppose $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. We will call f a *multivariate* function. We will commonly write,

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{pmatrix}$$

Example Functions

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}^2$,

$$f(t) = (t^2, \sqrt{t})$$

Example Functions

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as

$$f(r, \theta) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$$

Example Functions

Suppose we have some policy $\mathbf{x} \in \Re^M$. Suppose we have N legislators where legislator i has utility

$$U_i(\mathbf{x}) = \sum_{j=1}^M -(x_j - \mu_{ij})^2$$

We can describe the utility of all legislators to the proposal as

$$f(\mathbf{x}) = \begin{pmatrix} \sum_{j=1}^M -(x_j - \mu_{1j})^2 \\ \sum_{j=1}^M -(x_j - \mu_{2j})^2 \\ \vdots \\ \sum_{j=1}^M -(x_j - \mu_{Nj})^2 \end{pmatrix}$$

Jacobian

Definition

Suppose $f : X \rightarrow \mathbb{R}^n$, where $X \subset \mathbb{R}^m$, with f a differentiable function. Define the **Jacobian** of f at $\mathbf{x}$ as

$$\mathbf{J}(f)(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_m} \end{pmatrix}$$

Example of Jacobian

$$f(r, \theta) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}$$

$$\mathbf{J}(f)(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$